

# Zachary Metcalf

• Email: [zachmetcalf@gmail.com](mailto:zachmetcalf@gmail.com) • LinkedIn: [www.linkedin.com/in/zachmetcalf/](http://www.linkedin.com/in/zachmetcalf/) •

## Industry Experience

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### Riot Games

#### Manager, Software Engineering

Spr. 2026—Pres.

VALORANT—Competitive 5v5 character-based tactical shooter (Unreal Engine, C++)

- <http://www.playvalorant.com>
- Managed teams of up to 5 engineers across hiring, performance management, and career growth
- Led technical direction and hands-on implementation for major projects improving developer iteration

#### Manager, Software Engineering

Fall 2023—Spr. 2026

R&D—R&D (Unreal Engine, C++)

- Championed performance analytics, reference analysis, and debug tooling across multi-game ecosystem
- Led technical and product direction for cross-discipline maps, ai, audio, and performance pod
- Implemented and contributed modular Slack Api plugins to multi-game ecosystem

#### Senior Software Engineer

Spr. 2021—Fall 2023

R&D—R&D (Unreal Engine, C++)

- Built a playtest service for rapidly patching, deploying, and managing prototype builds on AWS servers
- Wrote a suite of C++ libraries for creating bespoke prototype tooling applications using ImGui
- Prototyped game features including: Wwise footsteps audio, services in Go, and a Slackbot in Python

### Blizzard Entertainment

#### Software Engineer

Spr. 2019—Spr. 2021

Overwatch—Colorful team-based hero shooter with exhilarating competitive 6v6 combat (C++)

- <http://www.playoverwatch.com>
- Produced, designed, and implemented features for the Workshop, a custom game scripting system
- Developed a framework for creating and extending synchronized debug widgets using ImGui
- Optimized asset package loading to support intelligent preloading and accommodate rapid hero swaps

### Rockstar San Diego

#### Engine Programmer

Sum. 2015—Fall 2018

Red Dead Redemption 2—Vast and atmospheric open-world tale of America's unforgiving heartlands (C++)

- <http://www.rockstargames.com/reddeadredemption2/>
- Developed population system to manage streaming, memory, and design requirements of the world
- Integrated proprietary bug tracking system for automated error detection and bug generation by QA
- Added debug tooling to visualize and auto-document game data for streaming, memory, and ped systems

## Academic & Teaching Experience

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### USC Department of Information Technology Program

#### Adjunct Lecturer

Fall 2022—Spr. 2023

ITP 435—Professional C++

- Applications of advanced concepts in C++ including lambda expressions, templates, secure coding, parallel programming, writing performant code, CMake and continuous integration

## University of Southern California

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M.S. in Computer Science

Sum. 2015

B.S. in Electrical Engineering

Spr. 2015

B.S. in Business Administration

Fall 2014